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Estimation of transmission distance between cases of (re-)emerging respiratory infectious diseases and its potential application in outbreak response

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A B S T R A C T

Quantifying transmission distance helps to understand infectious disease spread patterns, but few studies have assessed this for (re-)emerging respiratory infectious diseases. The COVID-19 pandemic's extensive surveillance and high-quality data in China offer a strong model for studying the spread pattern of respiratory infections. We collected COVID-19 outbreak data from June 2020 to December 2022 in mainland China to estimate transmission distances, examine the association between urban characteristics and transmission distance, and explore its implications in outbreak response. The mean transmission distance of all outbreaks was 18.94 km. This distance varied by SARS-CoV-2 variants, with wild-type outbreaks showing shortest transmission distance (3.08 km) and Omicron outbreaks showing longest transmission distance (20.17 km). Transmission distance was positively associated with city size and total length of roads. Potential transmission areas identified as areas within transmission distance of the first 100 cases contained 63.84–99.61% of cases. Providing testing services to residents in these areas helped to control the outbreak. Our results suggest that transmission distances can help predict where emerging respiratory infectious diseases may spread next spatially and guide the implementation of targeted interventions.

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1. Introduction

The mean transmission distance is the average of the geographical distances between infector-infectee pairs (e.g. by residential area or by point of reporting) and is a representation of the transmissibility of a pathogen and the spatial connectivity in a given population. The mean transmission distance may be used to predict where the disease may spread next spatially. With a short mean transmission distance, spatial clustering may occur (Salje, Cummings, & Lessler, 2016). With information on mean transmission distance, public health decision-makers could mitigate the spread of an outbreak by prioritising areas most likely affected. Factors such as demographics, human mobility, social contact patterns, city characteristics, and pathogen characteristics may all affect mean transmission distance (Brockmann & Helbing, 2013; Mao & Bian, 2010; Zhang et al., 2020). Understanding the mean transmission distance of infectious diseases and how their association with underlying contributors are valuable to future epidemic and pandemic preparedness.

Non-pharmaceutical interventions (NPIs), such as physical distancing, may be implemented to slow down the transmission of respiratory pathogens and have been widely implemented during the COVID-19 pandemic (Talic et al., 2021). However, these NPIs may have substantial negative impacts on social life, economic activities, and the mental health of the population (TheLancet Infectious Diseases, 2020; Bonaccorsi et al., 2020; Fezzi & Fanghella, 2021). Therefore, it is particularly valuable to find ways to manage outbreaks early on while the affected areas are still limited (Riley, 2007). Effective NPIs in targeted areas prevent the further spread of infectious diseases and limit the negative impacts of having to implement NPIs widely and prolongedly down the line. Therefore, estimating the transmission distance and subsequently identifying the potentially affected areas play a crucial role in understanding the spread of (re-)emerging respiratory infectious diseases.

Estimating the mean transmission distance in an outbreak is conceptually simple yet technically challenging. If all infector-infectee pairs can be clearly identified, the mean transmission distance is simply the distance between the two individuals involved. In practice, most infector-infectee pairs (i.e. the who-infected-whom relationships) cannot be observed during outbreak investigation. In rare cases when they are, their geographic coordinates are not always known.

During the COVID-19 pandemic in mainland China, mass and repeated testing (using nucleic acid amplification tests (NAATs)) has played a crucial role in containing the spread of SARS-CoV-2 and the case detection rates have therefore been exceptionally high (Li et al., 2021). Before the Omicron wave at the end of 2022, detailed outbreak investigations were carried out for all local outbreaks, generating valuable data that could be used to estimate the mean transmission distance of SARS-CoV-2. In this study, we used data on COVID-19 outbreaks in mainland China from June 2020 to December 2022 to estimate the mean transmission distance of SARS-CoV-2. We then investigated the association between mean transmission distances and their potential determinants.

2. Methods

2.1. Data collection

China's "zero-COVID" policy largely kept SARS-CoV-2 at bay for the general population before December 2022. There were sporadic and local COVID-19 outbreaks in many cities after the first wave of the pandemic was controlled in March 2020. However, none of them led to national-level transmission. In this study, a COVID-19 outbreak was defined as local transmissions that occurred after the introduction in a city with no COVID-19 cases previously. We included outbreaks that met the following criteria: (a) occurred in mainland China between April 1, 2020 and December 31, 2022, when we could assume that all infections were identified through mass testing programmes; (b) lasted for more than 10 days; (c) reported more than 100 infections cumulatively; and (d) had information on reporting date and residential address for each infection. We obtained this information from the city-level Health Commission/Center for Disease and Prevention (see Table S1, archived at Zenodo) (Zhu, 2023).

Data on SARS-CoV-2 variants for each outbreak were based on reports from city-level Health Commission/Center for Disease Control and Prevention, research articles, and news sources (Table S2) (Liu et al., 2021; Tan et al., 2020; Wang et al. 2022, 2023; Zhang et al., 2022; Zhu, 2023). In cases where variant information was not available, we assumed the outbreak was driven by the same variant circulating in mainland China during the same period. We considered four SARS-CoV-2 variants: wild-type, Alpha, Delta, and Omicron. Generation time distribution used for each variant was based on a systematic review (Table S3) (Xu et al., 2023).

Data on city characteristics, including population size, city size, total lengths of major urban roads, total numbers of bus stations, and daily intra-city travel intensity index were collected to analyze their association with mean transmission distance. Further details on outbreak definition and data collection can be found in the **Supplementary Methods** (Sina Finance, 2022; He, Lau, et al., 2020; Kretzschmar et al., 2020; Salje, Cummings, & Lessler, 2016; Song et al., 2022; Zhujiang Times, 2022).

2.2. Estimating mean transmission distances between cases

Previous studies indicated that interpersonal contacts primarily occurred within households, which served as main venues for the transmission of respiratory infectious diseases (Zheng et al., 2022; Zhu, Yuan, et al., 2023). Therefore, this study assumed that infections occurred within or near their homes, using residential addresses to estimate transmission distances. Using the generation time distributions and reported infections, we constructed the Wallinga-Teunis matrices (Salje, Cummings, & Lessler, 2016; Wallinga & Teunis, 2004). The probability ($P(\theta,t_1,t_2,I_{t_1},I_{t_2},\mathbb{C})$) that a case ($I_{t_1}$) occurring at the time t_1 linked to a case (I_{t_2}) at the time t_2 by θ transmission events in a potential transmission chain ($\mathbb{C}$) was estimated using the Wallinga-Teunis matrices. When the number of cases is small, we could exhaustively reconstruct possible transmission events linking infector-infectee pairs. The mean transmission distance can then be calculated using the distances between pairs, weighted by their probabilities (i.e. $P(\theta,t_1,t_2,I_{t_1},I_{t_2},\mathbb{C})$. When the number of cases is large, exhaustively reconstructing all possible transmission events is no longer computationally feasible. Therefore, we used the approach proposed by Salje et al. to approximate the overall mean transmission distance through Monte Carlo sampling with 1000 iterations (Salje, Cummings, & Lessler, 2016). The equations and additional information about estimating the mean transmission distance can be found in the **Supplementary Methods** and Fig. S1 (Sina Finance, 2022; He, Lau, et al., 2020; Kretzschmar et al., 2020; Salje, Cummings, & Lessler, 2016; Song et al., 2022; Zhujiang Times, 2022).

2.3. Agent-based model and simulation scenarios

An agent-based model (ABM) with spatial dimensions was developed to demonstrate the applied value of mean transmission distance. The agent had attributes such as disease statuses (infected/exposed, detectable, symptom onset, or isolated), time statuses (such as incubation period and generation time), spatial statuses (location coordinates and transmission distance), etc (Table S4). The number of secondary cases generated from each infection follows a negative binomial distribution with a mean equal to the basic reproduction number (R_0), which was 3.20 in the main analysis, and an overdispersion parameter k of 0.10 (Endo et al., 2020; Zhu, Wen, et al., 2023). The distance between infectors and their infectees follows an exponential distribution with a mean equal the mean transmission distance of the wild-type variant we estimated (Giles et al., 2019). Each new case was assigned an exposure time from the generation time distribution (gamma distribution with mean = 4.95 and sd = 1.92) (Xu et al., 2023). Symptomatic infections would be isolated with the assumption that they could not generate onward transmission ($R_0 = 0$). During the outbreaks of emerging infectious diseases, population mobility, migration, and demographic shifts were considered to be negligible. Thus, we used a hypothetical population of 100,000 susceptible individuals over an area of 80 km × 80 km to illustrate the role of the mean transmission distance in outbreak response. Parameters in ABM used for main analysis can be found in Table S5 (Xu et al., 2023; Hellewell et al., 2020; Endo et al., 2020; Giles et al., 2019; He, Yi, & Zhu, 2020).

In the baseline scenario, no NPI was implemented. The potential transmission areas were defined as areas within the mean transmission distance of identified cases. We varied the potential transmission area by including the area with mean transmission distance to the first n identified cases where n was set to 10 to 150 with 10 increments.

We then considered a mass testing programme where NAAT services were provided to 90% of individuals in potential transmission areas. Individuals who tested positive were isolated. We calculated the effectiveness of this mass testing programme as indicated by the number of cumulative cases reduced. We varied the coverage of NAAT services between 10 and 90%, with 10% increments and the areas of implementation between 10 and 90% quantile of the transmission distance assuming exponential distribution, with 10% increments. The ABM was run 1000 times using each set of parameters. This conceptual framework has been further elaborated using an example in **Supplementary Methods** and Fig. S2 (He, Lau, et al., 2020; Kretzschmar et al., 2020; Salje, Cummings, & Lessler, 2016; Sina Finance, 2022; Song et al., 2022; Zhujiang Times, 2022).

2.4. Sensitivity analysis

A sensitivity analysis was conducted to evaluate the impact of the generation time distribution on the estimation of the mean transmission distance. To give the mean and standard deviation of the generation time the same changes, the gamma distribution was used, with the shape parameter held constant and the scale parameter was multiplied by a multiplier (ranging from 0.75 to 1.25 with 0.05 increments). The multiplier value of 1 represented the parameters used in the main analysis.

To explore the impact of the characteristics of different pathogens on the application of mean transmission distance in the testing programmes, two sets of parameters were used for sensitivity analyses in ABM (Table S6) (Giles et al., 2019; He, Yi, & Zhu, 2020; Xu et al., 2023; Zhu, Wen, et al., 2023). One sensitivity analysis considered the spread of highly infectious pathogens in susceptible populations with R_0 of 7.90 and mean transmission distance of the Omicron variant. Another sensitivity analysis assessed the effectiveness of testing programmes in outbreaks caused by the transmission of pathogens with high R_0 and long mean transmission distance in immune/vaccinated populations. Additionally, to simulate a more realistic scenario of isolation, it was assumed that infections under quarantine may also cause transmission ($R_0 = 0.5$). The parameters used in the sensitivity analyses can be found in Table S6 (Giles et al., 2019; He, Yi, & Zhu, 2020; Xu et al., 2023; Zhu, Wen, et al., 2023).

2.5. Statistical analysis

The mean transmission distance in a given outbreak was averaged from the distance between each infector-infectee pair observed in that outbreak. The uncertainty was expressed using 95% uncertainty ranges (95% UR), which was calculated using the mean value plus or minus the margin of error (Z-value of 2.5% and 97.5% percentage points of the normal distribution multiplied by standard deviation). The mean transmission distance by variant was averaged from the mean transmission distance of all outbreaks driven by that variant. The uncertainty was expressed using the 25 percentile and 75 percentile values (P25, P75).

Mann-Kendall trend test was used to assess the changes in transmission distances by variant (wild-type, Alpha, Delta, and Omicron). Pearson correlation coefficients (r) were used to analyze the correlation between transmission distance and city characteristics. In order to control for variant characteristics, we only calculated the aforementioned Pearson correlation coefficients for outbreaks driven by the Omicron variants.

Friedman rank sum test was used to compare the mean transmission distance of each variant (Alpha, Delta, and Omicron) estimated with different generation times, and the post hoc Nimenyi test was used to compare the difference of each pair sensitivity analysis parameter set. All statistical tests were two-sided, and a p-value less than 0.05 was considered statistically significant.

Transmission distances were estimated using MATLAB 2021a. The agent-based model, statistical tests, and all data visualisation were implemented using R software version 4.3.1.

3. Results

We gathered data on 48 COVID-19 outbreaks in 27 cities across mainland China between June 11, 2020 and November 21, 2022 (Table S7). Out of these outbreaks, 39 were driven by the Omicron variant, five by the Delta variant, three by the Alpha variant, and one by the wild-type virus. The mean transmission distance of all outbreaks was 18.94 km (95% Uncertainty Range (UR): 3.61–20.20 km). The only wild-type outbreak occurred in Beijing and had a mean transmission distance of 3.08 km (95% UR: 0.78–5.38 km) (Fig. 1). The Alpha outbreaks had an overall mean transmission distance of 10.27 km (25 percentile, 75 percentile (P25, P75): 6.15 km, 13.64 km), ranging from 3.51 km (95% UR: 1.38–5.65 km, Tonghua (Jilin province)) to 18.51 km (95% UR: 11.16–25.85 km, Shijiazhuang (Hebei province)). The Delta outbreaks had an overall mean transmission distance of 17.73 km (P25, P75: 3.00 km, 25.27 km), ranging from 2.85 km (95% UR: 1.43–4.27 km, Yangzhou (Jiangsu province)) to 54.23 km (95% UR: 0–117.05 km, Zhengzhou (Henan province)). The Omicron outbreaks had the highest overall mean transmission distance of 20.17 km (P25, P75: 4.16 km, 25.89 km), ranging from 1.74 km (95% UR: 0–4.55 km, Shenzhen (Guangdong province)) to 93.46 km (95% UR: 3.54–183.38 km, Tianjin). The mean transmission distance increased as the pandemic progressed (i.e. from wild-type to Omicron outbreaks) (Mann-Kendall trend test, $p = 0.09$). In cities where multiple outbreaks occurred, driven by different variants, Omicron outbreaks had longer mean transmission

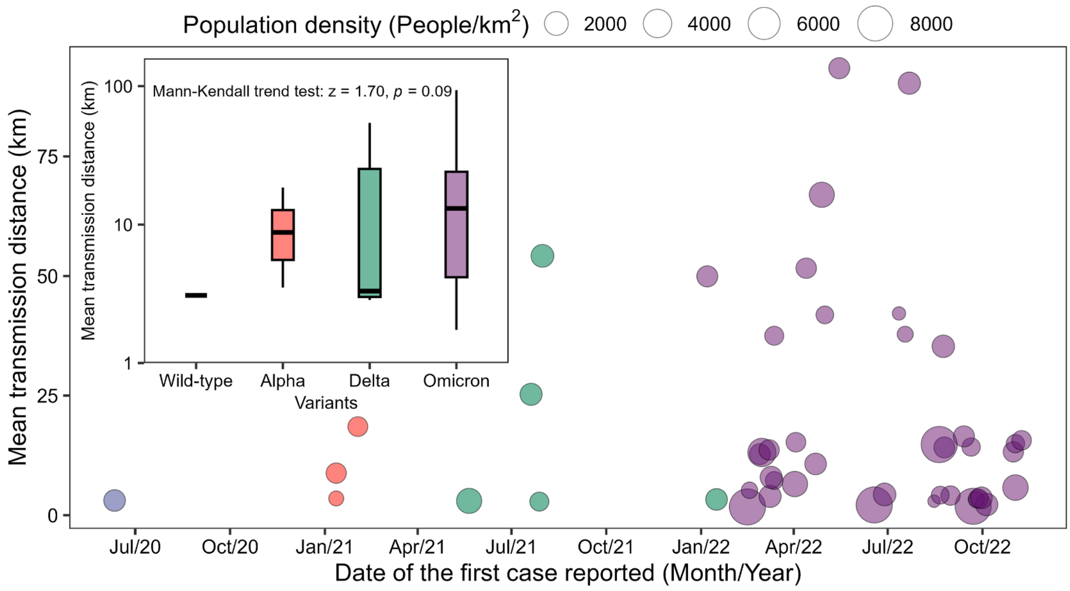

Fig. 1. The mean transmission distance of 48 COVID-19 outbreaks occurred between June 11, 2020 and November 21, 2022 in 27 cities in mainland China. Each point represents an outbreak. The point size represents the population density where the outbreak took place. The point colour represents variants of concern detected. The embedded plot shows the distribution of transmission distance by variant.

distances (see Fig. S3). For example, the mean transmission distance of Omicron outbreaks was 2.17 times longer in Beijing and 12.25 times longer in Guangzhou City, Guangdong, compared to the Delta outbreaks that previously occurred in the same cities.

Pearson correlation coefficients (r) were calculated to assess the association between city characteristics and mean transmission distance (Fig. 2). To control for the variants, only 34 outbreaks of Omicron variants in 19 cities were included in this analysis. There was a significant positive correlation between mean transmission distance and city size ($r = 0.46$, $p < 0.01$). The correlation between mean transmission distance and total length of roads was positive but not statistically significant ($r = 0.27$, $p = 0.12$). There was a significant negative correlation between mean transmission distance and population density ($r = -0.34$, $p = 0.05$).

In assessing the influence of the generation time distribution on estimating the mean transmission distance, we employed the gamma distribution with the shape parameter held constant, while the scale parameter was adjusted using a

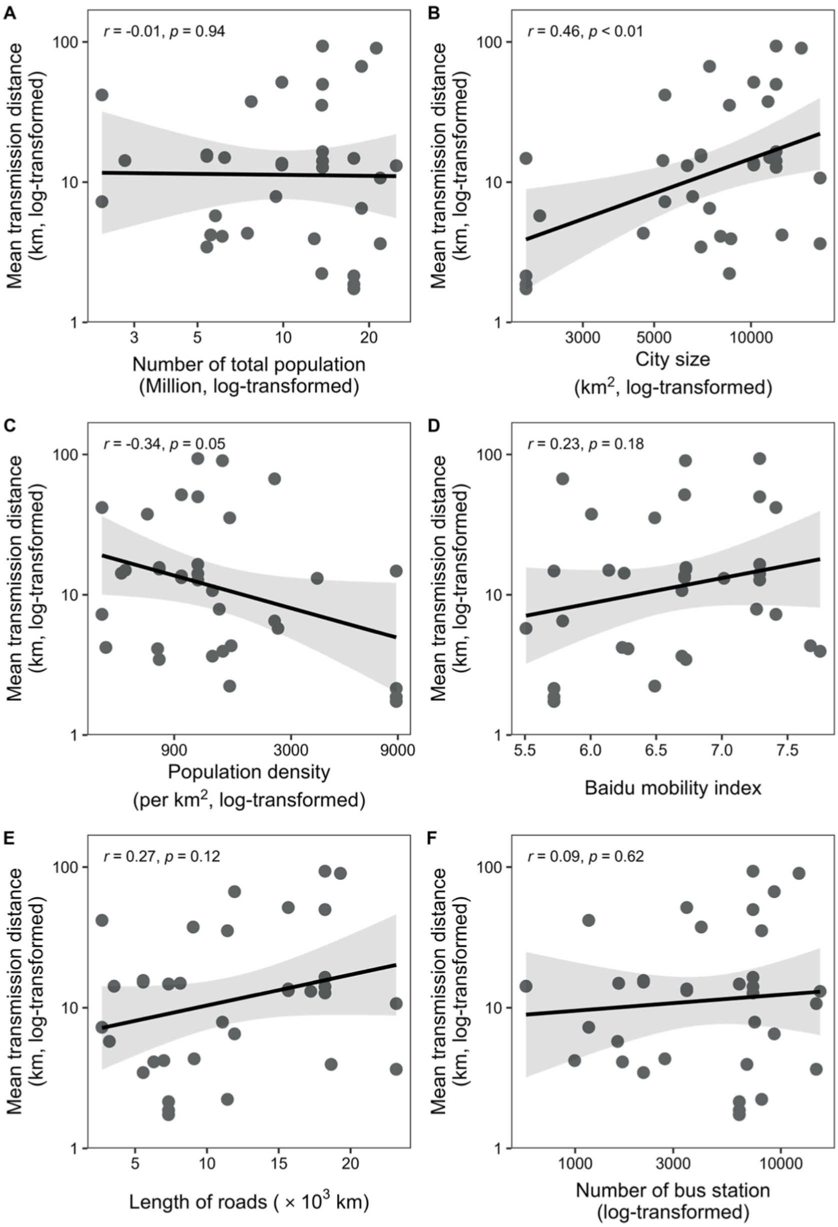

Fig. 2. The association between city characteristics and mean transmission distance among 34 outbreaks of the Omicron variant in 19 cities in mainland China. The solid lines are fits to univariate linear regression models; the shaded areas in **A–F** represent 95% confidence intervals. The Pearson's correlation coefficients (r) and the corresponding p-values are presented in the top left. Variables (mean transmission distance, number of total population, city size, population density, and number of bus stations) that are not normally distributed are log10-transformed.

multiplier ranging from 0.75 to 1.25. A higher generation time led to longer estimated mean transmission distances (Mann-Kendall trend test, $p < 0.01$, Fig. S4). In outbreaks driven by the Alpha and Delta variants, there was no significant difference between the main analysis (multiplier = 1) and the other multiplier analyses (varying between 0.75 and 1.25, Fig. S5). However, in the case of the Omicron variant, the mean transmission distance of the main analysis was significantly longer than those estimated with small multipliers (0.75, 0.80, and 0.85, $p < 0.01$), but significantly shorter than those estimated with large multipliers (1.15, 1.20, and 1.25, $p < 0.01$). These results indicated that the transmission distance of the Omicron variant was more sensitive to mean generation time than other variants.

The results of ABM indicate that an outbreak caused by a pathogen with $R_0 = 3.20$ would result in 44,426 cases (P25, P75: 33,494, 54,011) without intervention. To demonstrate the early warning potential of mean transmission distance in outbreak control, we defined potential transmission areas within 3.08 km of the first 10 to 150 cases and estimated the proportion of cases in these areas (Fig. 3). The proportion of cases in the potential transmission areas increased from 28.87% (95% CI: 27.11%, 30.62%) to 72.57% (95% CI: 70.88%, 74.27%) as the number of cases used to define these areas increased from 10 to 150. When the potential transmission areas were defined with more than 80 cases, the proportion of cases in these areas was more than 60%. When the number of cases used to construct potential transmission areas increased from 10 to 20, the final proportion of cases in these areas increased by 9.58% (95% CI: 8.38%, 10.79%). However, when the cases used to define potential transmission areas increased from 140 to 150, the proportion of cases in these areas increased by only 0.95% (95% CI: 0.67%, 1.23%).

We further analysed the impact of testing programmes (with NAAT, with subsequent isolation after positive results) in the potential transmission areas (Fig. 4). If NAAT were provided to 90% of residents in areas within 3.08 km of a case, the cumulative number of cases would be reduced to 25,108 (95% CI: 20,006, 29,782), and the peak number of daily cases would be reduced by 44.26% (P25, P75: 41.93%, 46.28%) (Fig. 4A). Even if NAAT were provided to only 30% of residents in these areas, the cumulative number of cases could still be reduced by 15.39% (P25, P75: 14.10%, 16.58%) (Fig. 4B). The cumulative number of cases decreases with the increasing area size of NAAT provision (Fig. 4C). However, the margin of decrease in the cumulative number of cases was small with increasing area size.

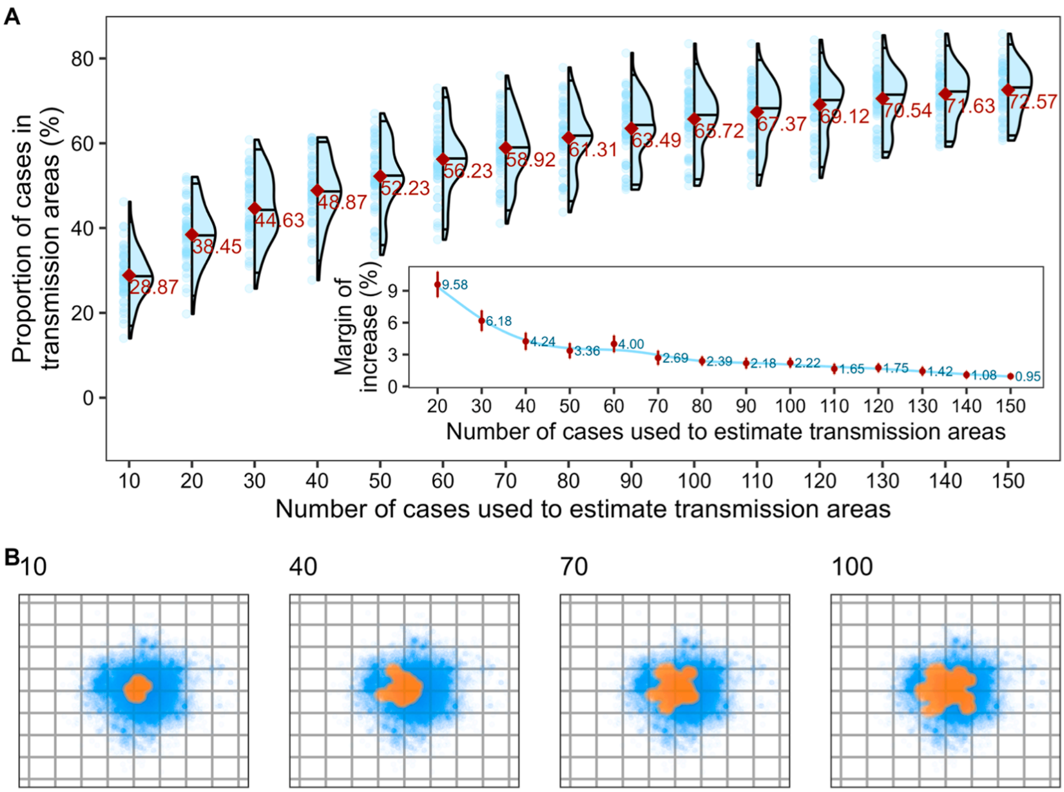

Fig. 3. The proportion of cases in the potential transmission areas (A), and geographic distribution of potential transmission areas (B) of an outbreak with $R_0 = 3.20$ and a mean transmission distance of 3.08 km. A: the red dots are the mean proportion; the inner plot is the margin of increase of the proportion, and the error bars are the 95% CIs. **B**: individuals were equally distributed in an area of 80 km * 80 km square (comparable to the area of Shanghai or Nanjing); the orange areas are the cases in the potential transmission areas; the potential transmission areas are estimated with a different number of cases (the number in the top left) in the early stage of the outbreak.

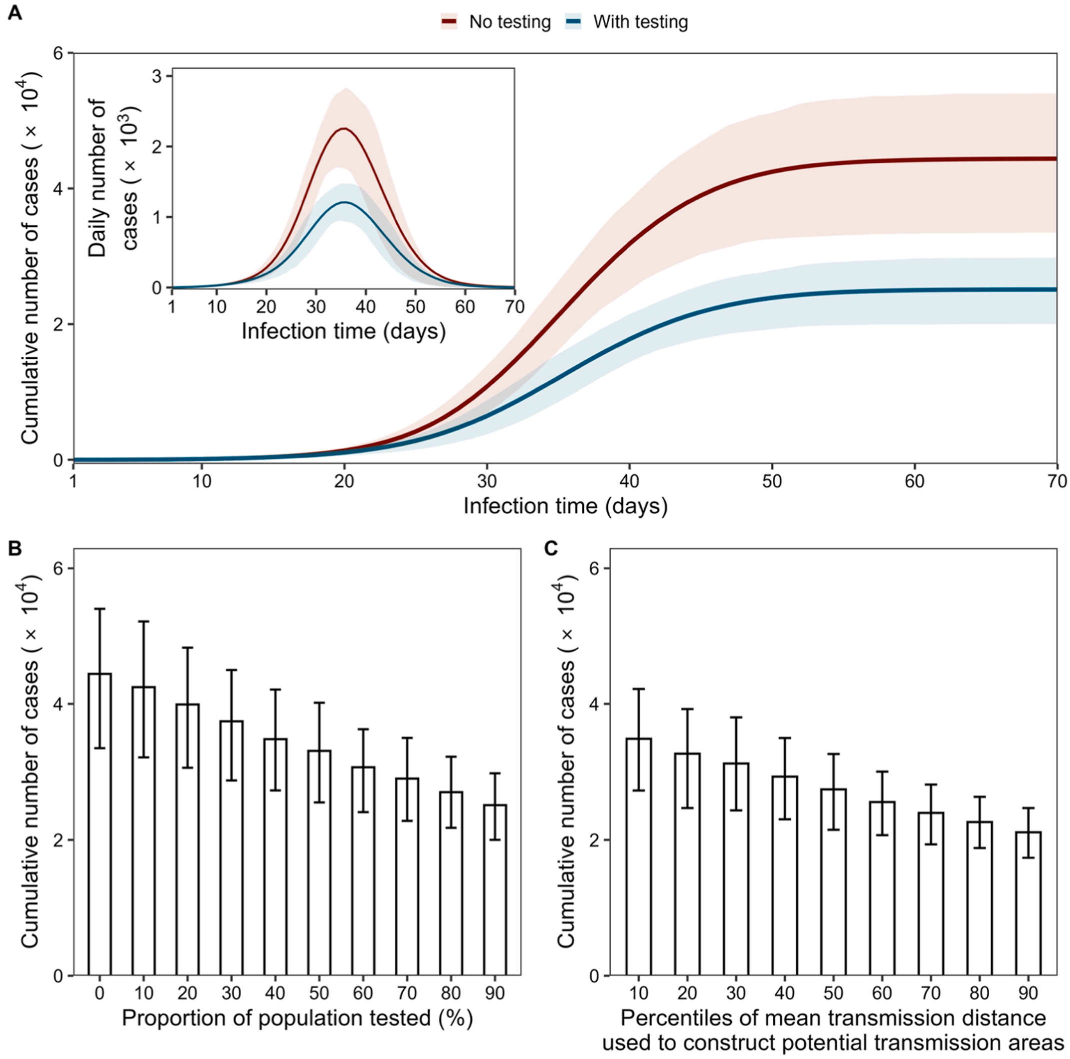

Fig. 4. (A) Epidemic curves and cumulative cases where 90% of the population within 3.08 km radius of each case identified were tested; (B) cumulative cases where 0–90% of the population within 3.08 km radius of each case identified were tested; (C) cumulative cases where 90% of the population within 0.32 (10 Percentile)–7.09 (90 Percentile) km radius of each case identified were tested. Mean reproduction number is 3.20. Uncertainties (25 percentile, 75 percentile) have been expressed as shaded areas in (**A**) and error bars in (**B**) and (**C**).

To explore the application of transmission distance in different scenarios, we varied R_0 and transmission distance for simulation in sensitivity analyses. Firstly, outbreaks caused by high transmissibility pathogens ($R_0 = 7.9$ and mean transmission distance = 20.17 km) were modeled. In the case, potential transmission areas constructed using longer transmission distances (20.17 km) contained more cases. Even the potential transmission areas identified with 10 cases contained 60.42% (95% CI: 58.36%, 62.48%) of the cases (Fig. S6). If NAAT were provided to 90% of residents in potential transmission areas (within 20.17 km of the cases), the cumulative number of cases would be reduced by 45.78% (P25, P75: 40.22%, 49.66%) (Fig. S7). Secondly, we simulated the spread of highly infectious pathogens in vaccinated populations with R_0 of 3.20 and mean transmission distance of 20.17 km (Fig. S8 and S9). The proportions of cases in potential transmission areas in this sensitivity analysis were higher than those in the main analysis, however, their changes were similar (Figure S8 and Fig. 3). If 90% of the population residing within a radius of 20.17 km from each case were tested, the decrease in the cumulative cases in the sensitivity analysis (44.09%; P25, P75: 40.54%, 47.90%) paralleled that in the main analysis (42.70%; P25, P75: 39.89%, 45.27%). In line with the main analysis ($R_0 = 3.2$ and mean transmission distance = 3.08 km, Fig. 4), the larger the NAAT area and the larger the proportion of people tested, the lower the cumulative number of cases in both sensitivity analyses (Fig. S7 and S9). Thirdly, we simulated a scenario in which transmission occurs even when infections were isolated (Fig. S10 and S11). The results of this scenario were almost identical to those of the primary analysis, namely that transmission during isolation had a negligible impact on the findings of this study.

4. Discussion

This study estimated the mean transmission distance for SARS-CoV-2 in mainland China. Our findings revealed that the mean transmission distance of Omicron outbreaks (20.17 km; P25, P75: 4.16 km, 25.89 km) was comparable to that of Delta outbreaks (17.73 km; P25, P75: 3.00 km, 25.27 km), and both were longer than those of Alpha outbreaks (10.27 km; P25, P75: 6.15 km, 13.64 km) and wild-type outbreaks (3.08 km, 95% CI: 0.78-5.38 km). The Omicron variant has the highest R_0 (of 7.9) among these four SARS-CoV-2 variants, while the wild-type has the lowest ($R_0 = 3.2$) (Liu & Rocklov, 2021, 2022; Perez-Guzman et al., 2023; Zhu, Wen, et al., 2023). Our results suggest that highly transmissible pathogens may also spread over longer distances in space (Zhu, Wen, et al., 2023).

Our results showed that mean transmission distance was positively associated with intra-city mobility, city size, and total length of roads and significantly negatively associated with population density. The long intra-city travel/commuting distance and well-developed public transportation in large cities may contribute to the far spread of infectious diseases (Rao et al., 2021). These results contribute to improving our understanding of the transmission of respiratory diseases in urban settings. Nevertheless, these associations may be influenced by unmeasured confounding factors, such as the use of public transport and the timing and stringency of local NPIs, and should not be taken as proof of causation. Future research is essential to further investigate the potential mechanisms underlying these associations.

Through model simulation, we demonstrated that the potential transmission area established around the first 100 cases contained about 63.84–99.61% of cases during the entire outbreak. Mean transmission distances could inform public health practitioners on where the pathogen may spread. We showed that mass testing areas in potential transmission areas identified using the mean transmission distance could reduce the cumulative cases and lower the epidemic peak sizes. The decrease in the cumulative cases was more sensitive to the proportion of the population tested than to the distance used to define potential transmission areas, suggesting the importance of high uptake rates of the testing programme. Although we only analysed the impacts of mass testing programmes, the planning of many other NPIs (e.g. physical distancing, contact tracing) may also benefit from spatial prioritisation informed by mean transmission distance in order to minimise social disruption (Zhu et al., 2022).

The study has some limitations. First, we did not consider the differences in terms of NPI implementation during each outbreak as such data on the local level was not available. This may have introduced bias as NPIs may affect human social contact and mobility patterns and alter disease transmission dynamics (Badr et al., 2020; Zhang et al., 2020). Stringent NPIs may restrict the mobility of individuals, reducing the transmission distance. Second, the infector-infectee relationships were assumed to be completely unobserved in this study and were all estimated using generation time. The uncertainty in the inference process may be reduced by incorporating contact tracing information. This study used residential addresses to estimate transmission distances, assuming infections occurred near their homes. Integrating information such as the workplaces and travel trajectories of infected individuals could improve the accuracy of these estimates. However, such information was unfortunately not available to us. Third, our data were derived from outbreaks under the "zero-COVID" policy, which ensured high data quality but featured uniquely stringent interventions and constrained population mobility. This limitation should be considered when applying the results to other areas. Future studies are needed to explicitly validate this approach in diverse settings where data quality, intervention intensity, and population mobility patterns differ.

In conclusion, the mean transmission distance of the SARS-CoV-2 Omicron/Delta variant was longer than that of the Alpha variant and the wild-type virus. City size and total length of roads were positively associated with the mean transmission distance. Estimating the transmission distance could help policy-makers plan for spatial prioritisation of their response efforts. Implementing interventions in areas identified using mean transmission distances may allow public health practitioners utilize their limited resources while minimizing social/economic disruptions while curbing the spread of emerging infectious diseases.

CRediT authorship contribution statement

Wenlong Zhu: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Xueyao Liang:** Writing – review & editing, Investigation, Data curation. **Xinyu Wang:** Writing – review & editing, Investigation, Data curation. **Wenyong Zhou:** Writing – review & editing, Investigation, Data curation. **Qianyi Ge:** Writing – review & editing, Investigation, Data curation. **Minyi Yang:** Writing – review & editing, Investigation, Data curation. **Bo Zheng:** Writing – review & editing, Investigation, Data curation. **Zexuan Wen:** Writing – review & editing, Investigation, Data curation. **Zhixi Liu:** Writing – review & editing, Investigation, Data curation. **Ye Yao:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Yang Liu:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Conceptualization. **Weibing Wang:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Ethics approval and consent to participate

All data were collected from publicly available sources. Data have been de-identified before collection, and the need for informed consent was waived.

Consent for publication

Not applicable.

Availability of data and materials

Data used in the study are publicly available from the sources listed in Supplementary Material Table S1 and S2, and Ref (Zhu, 2023). The codes used in the study are available on GitHub at https://github.com/ZhuWenlong95/CovidTransmissionDistance.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Not applicable.

List of abbreviations

NPIs	Non-pharmaceutical interventions
NAATs	Nucleic acid amplification tests
ABM	Agent-based model
R_0	Basic reproduction number
95% UR	95% uncertainty range
P25, P75	25% percentile and 75% percentile

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.idm.2026.02.001.

References

Badr, H. S., Du, H., Marshall, M., Dong, E., Squire, M. M., & Gardner, L. M. (2020). Association between mobility patterns and COVID-19 transmission in the USA: A mathematical modelling study. *The Lancet Infectious Diseases, 20*, 1247–1254.
Bonaccorsi, G., Pierri, F., Cinelli, M., Flori, A., Galeazzi, A., Porcelli, F., Schmidt, A. L., Valensise, C. M., Scala, A., Quattrociocchi, W., & Pammolli, F. (2020). Economic and social consequences of human mobility restrictions under COVID-19. *Proc Natl Acad Sci U S A, 117*, 15530–15535.
Brockmann, D., & Helbing, D. (2013). The hidden geometry of complex, network-driven contagion phenomena. *Science, 342*, 1337–1342.
Endo, A., Covid-Working Group Centre for the Mathematical Modelling of Infectious Diseases, Abbott, S., Kucharski, A. J., & Funk, S. (2020). Estimating the overdispersion in COVID-19 transmission using outbreak sizes outside China. *Wellcome Open Research, 5*, 67.
Fezzi, C., & Fanghella, V. (2021). Tracking GDP in real-time using electricity market data: Insights from the first wave of COVID-19 across Europe. *European Economic Review, 139*, Article 103907.
Giles, J. R., Salje, H., & Lessler, J. (2019). The IDSpatialStats R package: Quantifying spatial dependence of infectious disease spread. *R Journal, 11*, 308–327.
He, X., Lau, E. H. Y., Wu, P., Deng, X., Wang, J., Hao, X., Lau, Y. C., Wong, J. Y., Guan, Y., Tan, X., Mo, X., Chen, Y., Liao, B., Chen, W., Hu, F., Zhang, Q., Zhong, M., Wu, Y., Zhao, L., … Leung, G. M. (2020). Temporal dynamics in viral shedding and transmissibility of COVID-19. *Nat Med, 26*, 672–675.
He, W., Yi, G. Y., & Zhu, Y. (2020). Estimation of the basic reproduction number, average incubation time, asymptomatic infection rate, and case fatality rate for COVID-19: Meta-analysis and sensitivity analysis. *Journal of Medical Virology, 92*, 2543–2550.
Hellewell, J., Abbott, S., Gimma, A., Bosse, N. I., Jarvis, C. I., Russell, T. W., Munday, J. D., Kucharski, A. J., Edmunds, W. J., Covid-Working Group Centre for the Mathematical Modelling of Infectious Diseases, Funk, S., & Eggo, R. M. (2020). Feasibility of controlling COVID-19 outbreaks by isolation of cases and contacts. *Lancet Glob Health, 8*, e488–e496.
Kretzschmar, M. E., Rozhnova, G., Bootsma, M. C. J., van Boven, M., van de Wijgert, J., & Bonten, M. J. M. (2020). Impact of delays on effectiveness of contact tracing strategies for COVID-19: A modelling study. *The Lancet Public Health, 5*, e452–e459.
Li, Z., Liu, F., Cui, J., Peng, Z., Chang, Z., Lai, S., Chen, Q., Wang, L., Gao, G. F., & Feng, Z. (2021). Comprehensive large-scale nucleic acid-testing strategies support China's sustained containment of COVID-19. *Nat Med, 27*, 740–742.
Liu, Y., & Rocklov, J. (2021). The reproductive number of the Delta variant of SARS-CoV-2 is far higher compared to the ancestral SARS-CoV-2 virus. *J Travel Med, 28*, 1–3.

Liu, Y., & Rocklov, J. (2022). The effective reproductive number of the Omicron variant of SARS-CoV-2 is several times relative to Delta. *J Travel Med, 29*, 1–4.
Liu, S., Yuan, S., Sun, Y., Zhang, B., Wang, H., Lu, J., Tan, W., Liu, X., Zhang, Q., Xia, Y., Lyu, X., Li, J., & Guo, Y. (2021). A COVID-19 Outbreak - Nangong City, Hebei Province, China, January 2021. *China CDC Wkly, 3*, 401–404.
Mao, L., & Bian, L. (2010). Spatial-temporal transmission of influenza and its health risks in an urbanized area. *Computers, Environment and Urban Systems, 34*, 204–215.
Perez-Guzman, P. N., Knock, E., Imai, N., Rawson, T., Elmaci, Y., Alcada, J., Whittles, L. K., Thekke Kanapram, D., Sonabend, R., Gaythorpe, K. A. M., Hinsley, W., FitzJohn, R. G., Volz, E., Verity, R., Ferguson, N. M., Cori, A., & Baguelin, M. (2023). Epidemiological drivers of transmissibility and severity of SARS-CoV-2 in England. *Nature Communications, 14*, 4279.
Rao, Y. X., Yang, J. X., Dai, D. Y., Wu, K. S., & He, Q. S. (2021). Urban growth pattern and commuting efficiency: Empirical evidence from 100 Chinese cities. *Journal of Cleaner Production, 302*, Article 126994.
Riley, S. (2007). Large-scale spatial-transmission models of infectious disease. *Science, 316*, 1298–1301.
Salje, H., Cauchemez, S., Alera, M. T., Rodriguez-Barraquer, I., Thaisomboonsuk, B., Srikiatkhachorn, A., Lago, C. B., Villa, D., Klungthong, C., Tac-An, I. A., Fernandez, S., Velasco, J. M., Roque, V. G., Jr., Nisalak, A., Macareo, L. R., Levy, J. W., Cummings, D., & Yoon, I. K. (2016). Reconstruction of 60 years of Chikungunya epidemiology in the Philippines demonstrates episodic and focal transmission. *The Journal of Infectious Diseases, 213*, 604–610.
Salje, H., Cummings, D. A. T., & Lessler, J. (2016). Estimating infectious disease transmission distances using the overall distribution of cases. *Epidemics, 17*, 10–18.
Sina Finance. (2022). Why did Guangzhou undergo a series of large-scale nucleic acid tests during this epidemic?. https://finance.sina.com.cn/china/gncj/2022-04-14/doc-imcwipii4220159.shtml.
Song, C., Pei, T., Wang, X., Liu, Y., Jia, M., & Zhou, D. (2022). Dynamic characteristics of the COVID-19 epidemic in China's major cities. *Annals of GIS, 28*, 445–456.
Talic, S., Shah, S., Wild, H., Gasevic, D., Maharaj, A., Ademi, Z., Li, X., Xu, W., Mesa-Eguiagaray, I., Rostron, J., Theodoratou, E., Zhang, X., Motee, A., Liew, D., & Ilic, D. (2021). Effectiveness of public health measures in reducing the incidence of covid-19, SARS-CoV-2 transmission, and covid-19 mortality: Systematic review and meta-analysis. *BMJ, 375*, Article e068302.
Tan, W., Niu, P., Zhao, X., Pan, Y., Zhang, Y., Chen, L., Zhao, L., Wang, Y., Wang, D., Han, J., Gao, G. F., Huang, C., Xu, W., & Wu, G. (2020). Reemergent Cases of COVID-19 - Xinfadi Wholesales Market, Beijing Municipality, China, June 11, 2020. *China CDC Wkly, 2*, 502–504.
The Lancet Infectious, Diseases. (2020). The intersection of COVID-19 and mental health. *The Lancet Infectious Diseases, 20*, 1217.
Wallinga, J., & Teunis, P. (2004). Different epidemic curves for severe acute respiratory syndrome reveal similar impacts of control measures. *American Journal of Epidemiology, 160*, 509–516.
Wang, K., Luan, Z., Guo, Z., Ran, J., Tian, M., & Zhao, S. (2022). The association between clinical severity and incubation period of SARS-CoV-2 Delta variants: Retrospective observational study. *JMIR Public Health Surveill, 8*, Article e40751.
Wang, Y., Sun, K., Pan, Y., Yi, L., Huo, D., Wu, Y., Dong, S., Guo, J., Dou, X., Wang, W., Wu, S., Bai, X., Yu, H., & Wang, Q. (2023). A retrospective modeling study of the targeted non-pharmaceutical interventions during the xinfadi outbreak in the early stage of the COVID-19 pandemic - Beijing, China, 2020. *China CDC Wkly, 5*, 108–112.
Xu, X., Wu, Y., Kummer, A. G., Zhao, Y., Hu, Z., Wang, Y., Liu, H., Ajelli, M., & Yu, H. (2023). Assessing changes in incubation period, serial interval, and generation time of SARS-CoV-2 variants of concern: A systematic review and meta-analysis. *BMC Medicine, 21*, 374.
Zhang, J., Litvinova, M., Liang, Y., Wang, Y., Wang, W., Zhao, S., Wu, Q., Merler, S., Viboud, C., Vespignani, A., Ajelli, M., & Yu, H. (2020). Changes in contact patterns shape the dynamics of the COVID-19 outbreak in China. *Science, 368*, 1481–1486.
Zhang, X., Zhang, W., & Chen, S. (2022). Shanghai's life-saving efforts against the current omicron wave of the COVID-19 pandemic. *Lancet, 399*, 2011–2012.
Zheng, B., Zhu, W., Pan, J., & Wang, W. (2022). Patterns of human social contact and mask wearing in high-risk groups in China. *Infectious Diseases of Poverty, 11*, 69.
Zhu, W. (2023). Data for estimating the transmission distance. https://doi.org/10.5281/zenodo.10444974.
Zhu, W., Wen, Z., Chen, Y., Gong, X., Zheng, B., Liang, X., Xu, A., Yao, Y., & Wang, W. (2023). Age-specific transmission dynamics under suppression control measures during SARS-CoV-2 Omicron BA.2 epidemic. *BMC Public Health, 23*, 743.
Zhu, W., Yuan, S., Mao, S., Chen, J., Zheng, Y., Jiang, X., Yu, X., Jiang, C., Fang, Q., Wang, W., Yuan, Z., & Yao, Y. (2023). Relationship of close contact settings with transmission and infection during the SARS-CoV-2 Omicron BA.2 epidemic in Shanghai. *BMJ Global Health, 8*, Article e012289.
Zhu, W., Zhu, Y., Wen, Z., Zheng, B., Xu, A., Yao, Y., & Wang, W. (2022). Quantitative assessment of the effects of massive nucleic acid testing in controlling a COVID-19 outbreak. *BMC Infectious Diseases, 22*, 845.
Zhujiang Times. (2022). Guangzhou's latest report: 20 new local positive cases! announcement of virus gene sequencing results. https://new.qq.com/rain/a/20220429A092UN00.